

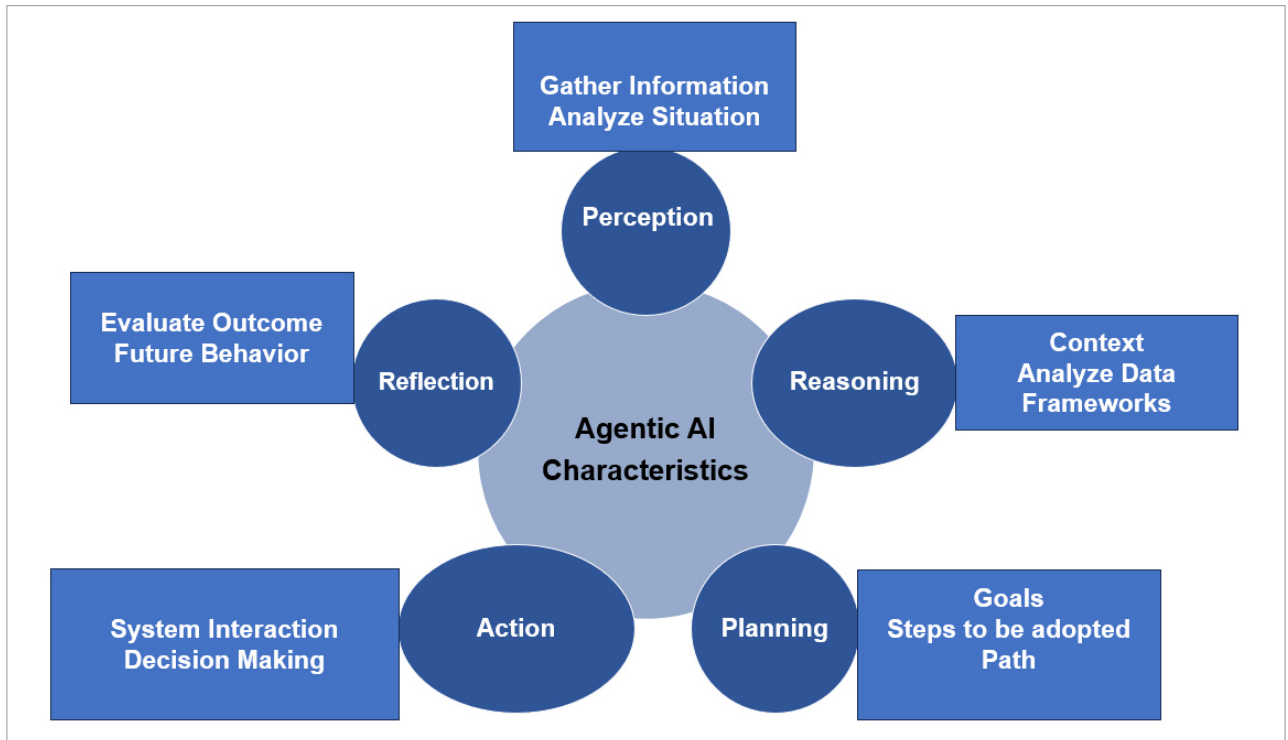


Figure 1: Agentic AI characteristics

Industry trends in agentic AI

- By 2027, nearly 15% of new applications will be automatically generated by AI without a human in the loop. This is not happening at all today. – *Gartner*
- Recent surveys indicate almost 80% of CEOs believe AI is likely to significantly enhance business efficiencies in their organisation. – *Forbes*
- According to a Deloitte survey, 42% of organisations already cite tangible benefits from AI agents.
- AI agents will lead to a flattening of organisational structures, with up to 20% of organisations eliminating middle management positions by 2026. – *Gartner*
- By 2029, agentic AI will autonomously resolve 80% of common customer service issues without human intervention, leading to a 30% reduction in operational costs. – *Gartner*
- By 2028, agentic AI will be responsible for making 15% of everyday work decisions. – *Forbes*

This continuous cycle of perception, planning, action, and reflection allows agentic AI to learn and improve over time.

Agentic AI reference architecture

Agentic AI systems follow single-agent and multi-agent architecture models. These systems leverage machine learning models and methods to enable intelligent decision making and automation.

Agentic AI architecture is robust, modular, and secure to support autonomous reasoning, workflow orchestration, and multi-agent collaboration. The architecture integrates planning, memory, execution, safety, and governance across the enterprise.

The various layers of the agentic AI architecture are classified as:

- Experience and channels layer
- Perception layer
- Cognitive / agent orchestration layer
- Action and tools layer
- Data and knowledge layer
- Integration layer
- Infrastructure and platform layer
- Operations and observability layer
- Governance, security, and risk layer

Experience and channels layer: This layer focuses on the user interface and how humans or external systems interact with the agents. Its components are:

- **User interfaces:** Web apps, portals, mobile apps, chat widgets, voice interfaces, CLI
- **Workplace add-ins:** Teams/Slack plugins, Outlook add-ins, intranet tiles, line-of-business UI embeds
- **System-to-system entry points:** Webhooks, event streams, scheduled jobs that trigger agents
- **Persona and role profiles:** Map enterprise roles to specific agents, tools, and permissions